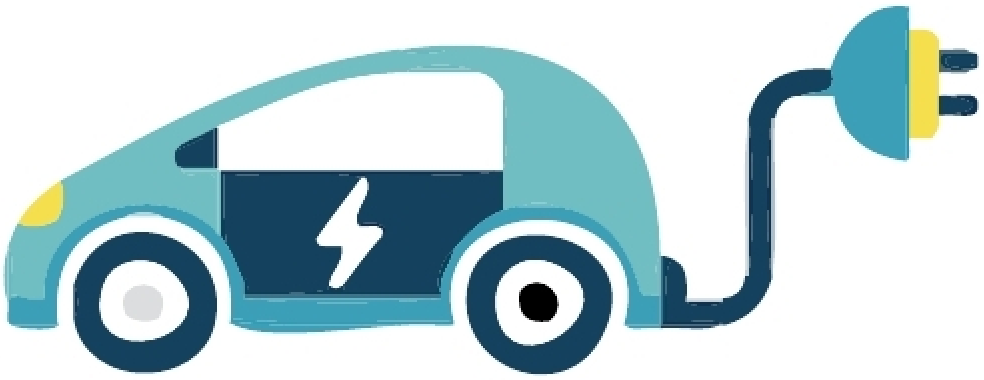

Jeetender Sharma, managing director, Okinawa Autotech Pvt Ltd

Q. What are your views on India's current EV ecosystem?

A. The global EV market is huge and India can cater to a good portion of that market, especially through two-wheelers. Last year, we sold 20 million vehicles in India. EVs will take over as a response to increasing petrol prices and pollution concerns.

In the next four to five years, the market will see a large number of EVs on the road. The government of India is aiming to bring five million EVs by 2022.

Vehicles being offered to consumers should have top quality, cost and delivery management. The price of EVs should eventually match that of internal combustion (IC) engines to pull consumers.

Q. What are the current challenges with EVs?

A. The most important challenge is lack of awareness about the benefits of EVs, in terms of economics, environment and performance.

Quality, cost and delivery (QCD) management is also a major area of focus. In fact, roads should be made smoother with reduced potholes and obstacles. This can ensure better vehicle performance going ahead.

Q. For consumers, what environmental benefits and return on investment can EVs bring?

A. It is not just the vehicle but the infrastructure that contributes to increasing environmental concerns. Roads are too narrow and the number of vehicles is too high. Traffic keeps increasing and in traffic, engines of the vehicles keep running, causing a large amount of polluted emissions. EVs can curb this carbon footprint significantly.

In terms of returns, our vehicles can run for ₹ 0.20 per km, as compared to petrol two-wheelers that require ₹ 1.80 per kilometre. Customers can reduce their fuel costs by almost ten-folds.

Moreover, our scooters consume two units of electricity for every 100 kilometres, essentially reducing the cost of charging. In fact, we have designed a battery for EVs that can be charged like a normal smartphone battery. This was innovated to address the queries of people regarding charging.

Given all calculations, we can say that compared to fuel-based vehicles, you can save up to ₹ 40,000 in two to three years with two-wheeler EVs like ours, essentially recovering the capital purchase cost.

Q. What are the difficulties faced by four-wheeler EVs as compared to two-wheeler ones?

A. The most notable challenge is the battery form factor. In four-wheelers, charging the battery is a matter of discussion. The weight and size make it difficult to charge the battery without a proper charging infrastructure.

In India, charging infrastructure for four-wheelers is not yet developed. Two-wheelers, on the other hand, have comparatively easily chargeable or replaceable batteries. Our proprietary battery technology has made it even more convenient.

Moreover, in case of four-wheelers, manufacturers are still figuring out a proper QCD management strategy. Indian consumers are used to IC vehicles. A shift to EVs will need adjustments. This will take its own course and time.

Q. How do you ensure quality and feature control of your electric scooters?

A. The design and sync between components is important. To ensure longevity at better speeds (equivalent to IC), we have added a few design elements. First is the inclusion of an electronic assisted braking system (EABS) that consumes the energy generated when the brake is applied. Energy is transferred and stored in the battery. Hence, it regenerates battery power and enhances battery life.

We have two models, one offers 100 kilometres, while the other offers 170 to 200 kilometres. The lithium battery pack we use can be charged to 80 per cent within 45 minutes. It has a capacity of 1.7kWh and weighs 9.5 kilograms in a compact form factor.

Q. How can traditional car producers revamp their manufacturing facility to produce EVs?

A. The first aim of car producers should be to identify their target market and predict demand. EV business must be started in a supply-and-demand-based model. The manufacturing facility requires an upgradation of the supply chain, including suppliers and the strategy.

Vehicle design needs to be changed, too. For instance, in EV, the motor, controller and battery pack constitute the complete engine. However, turning to EV reduces the overall bill of material by 30 to 40 per cent in the long run. The rest depends on the features and facilities the producer wants to provide. EFY